

多媒体技术 2026课程设计

2026/3/21

任务背景

- 生成技术发展同时带来了诸多现实风险



乌克兰总统泽连斯基宣布投降的“深度伪造”视频被疯传，社交媒体成为军事对抗的“第二战场”

国家安全风险



因AI换脸电信诈骗，某受害人10分钟内被骗走430万

经济安全风险



2022年10月初，
NovelAI 50G模型遭黑客泄露

知识产权风险

任务背景

- 从识别伪造内容到解释伪造痕迹

可疑内容

>>>

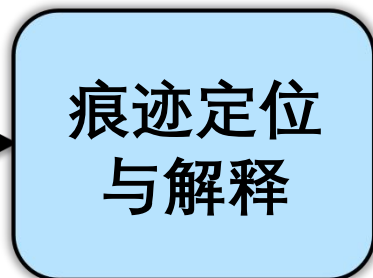
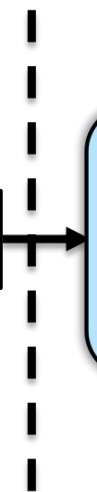
识别伪造内容

>>>

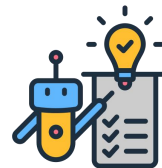
解释伪造痕迹



假



伪造解释



数据集

下载链接:

<https://huggingface.co/datasets/Coxy7/X-AIGD/tree/main/data>
(只需要labeled的两个文件)

• 数据集构成

- 真图: 4000 张, 来自SA-1B、LAION-Aesthetic、MSCOCO和Conceptual Captions
- 假图: 使用 13 个文本生成图像模型, 生成 52000 张图片。生成器覆盖 3 类架构 Diffusion Unet, Diffusion Transformer 和自回归模型
- 人工标注的痕迹: 3035 个有效标注, 其标注采用像素级分割掩码, 并将伪造痕迹划分为 3 个层级、7 个类别, 共包含 18,202 个伪影实例。

现有数据集对比

Dataset	Real Images	Artifact Annotation	
		Localization	#Category
PAL4VST (Zhang et al., 2023)	✗	Pixel mask	N/A
SynArtifact (Cao et al., 2024)	✗	Bounding box	13
RichHF-18K (Liang et al., 2024)	✗	Point	N/A
LOKI (image) (Ye et al., 2025)	Unpaired	Bounding box	N/A
FakeBench (Li et al., 2024a)	Unpaired	N/A	14
MMFR-Dataset (Gao et al., 2025)	Paired	N/A	5
SynthScars (Kang et al., 2025)	✗	Pixel mask	3
X-AIGD (ours)	Paired	Pixel mask	7

数据集

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• 伪造痕迹标注

- 底层 关注基础视觉属性: **Textures、Edges & Shapes、Color、Symbols**
- 高层 关注结构和组成是否合理: **Semantics**
- 认知层 关注常识与物理: **Commonsense、Physics**

Cognitive-level Counterfactuals

Violations of real-world knowledge, commonsense, or fundamental physical laws



Commonsense

Objects defying spatial, temporal, functional, or biological commonsense; linguistic errors



Physics

Violation of physical laws, e.g., inconsistent shadows and reflections disobeying optical laws

High-level Semantics

Semantic deviations due to issues of structural integrity or composition



Semantics

Objects with semantic errors or abnormal structure; incorrect or nonsensical composition of different parts of an object

Low-level Distortions

Distortions affecting fundamental visual properties and details



Edges & Shapes



Textures



Color



Symbols

数据集样例

- 图片数据
- 标注labels:
 - (痕迹分割+类型)

```
>>> print(df.loc[0])
image                {'bytes': b'\x89PNG\r\n\x1a\n\x00\x00\x00\rIHD...'}
generator            FLUX.1_dev-IPhonePhoto-raw
uid                  -3F5VKhT
labels               [{'label': 'low-level-symbol', 'points': [[ 70...
original_prompt      The image depicts a bustling urban intersectio...
positive_prompt       cinematic quality, ultra realistic, HDR, photo...
negative_prompt       Abstract, game, 3D style, fantastical, splash ...
guidance_scale        7.0
num_inference_steps  50.0
scheduler            FlowMatchEulerDiscreteScheduler
seed                  1
width                 1360
height                768
image_format          JPEG
jpeg_quality          85.0
chroma_subsampling    4:4:4
Name: 0, dtype: object
>>> print(df.loc[0].labels)
[{'label': 'low-level-symbol', 'points': array([array([ 70.25590551, 525.59055118]),
          array([ 72.22440945, 556.2992126 ]),
          array([122.22440945, 554.33070866]),
          array([120.05905512, 521.45669291])], dtype=object)},
 {'label': 'low-level-edge_shape', 'points': array([array([344.89057956, 427.75590551]),
          array([422.44963467, 428.1496063 ]),
          array([418.70947719, 413.18897638]),
          array([655.63063063, 390.08108108]),
          array([644.14414414, 377.24324324]),
          array([539.18191814, 386.22047244]),
          array([532.29215436, 379.33070866]),
          array([466.34727247, 378.54330709]),
          array([450.99294176, 385.23622047]),
          array([420.28428034, 390.94488189]),
          array([407.48900475, 387.00787402]),
          array([386.03231184, 394.68503937]),
          array([367.13467404, 403.74015748])])}]
```

评价标准

- **任务1（检测分类）**

- 汇报在真图+生成图像的分类准确率Acc, AP

- **任务2（伪造解释）**

- 汇报生成解释与标注解释的相似度（BLEU,）

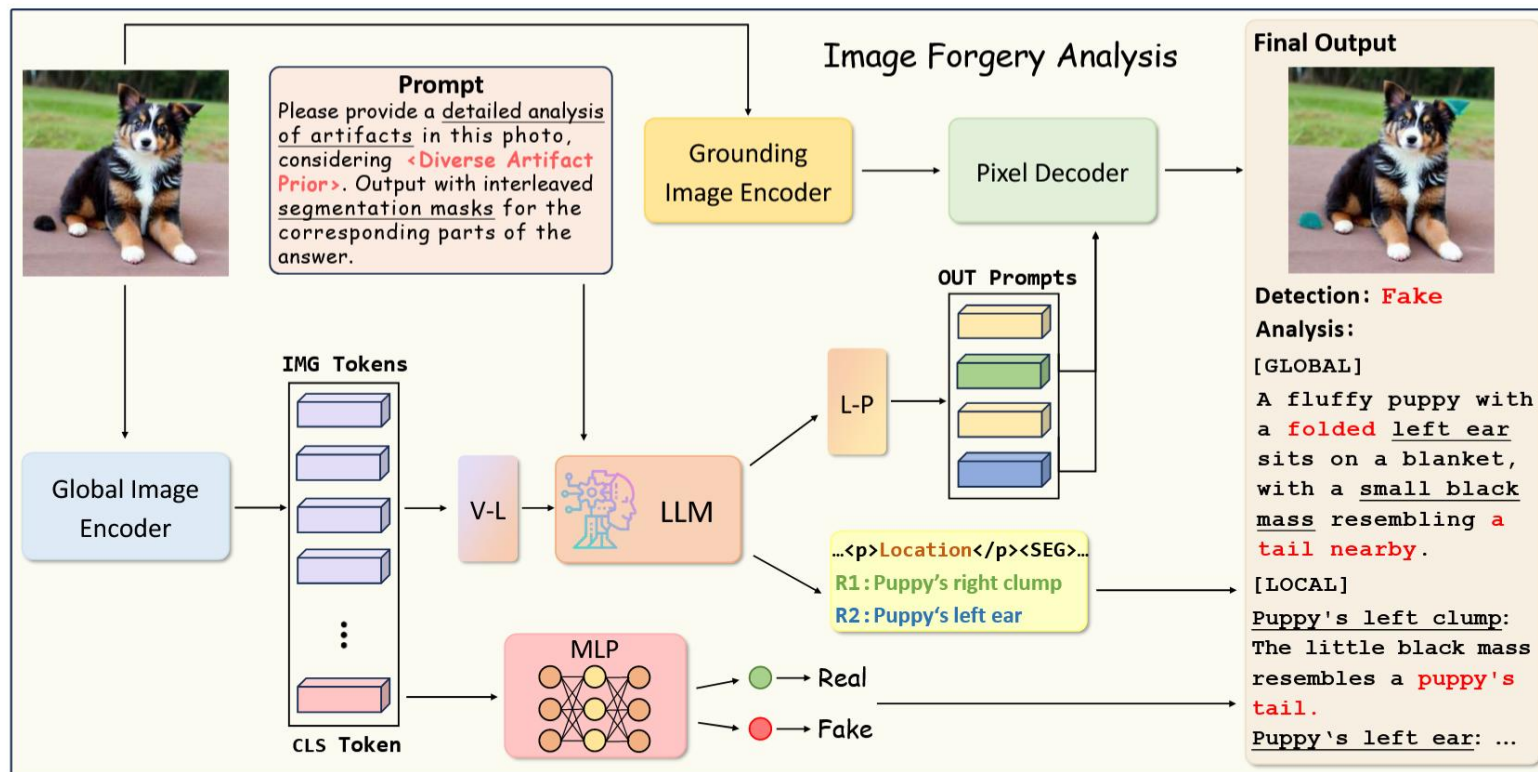
- **任务3（痕迹定位）**

- 汇报定位mask与标注区域的IoU、PixP、PixR、PixF1

检测已有工作示例--LEGION

LEGION: Learning to Ground and Explain for Synthetic Image Detection

- 同时支持检测、定位、文本解释的统一框架，并进一步把解释结果拿去反向指导图像修复与生成优化。



时间和要求

- 分组完成，自由组队，每组1-2人
- 形式：现场Slides答辩，每组须**全员**汇报，要求讲清来龙去脉和各自的工作量
- **答辩前提交代码包和Slides文件**
- 汇报时间：第八周
- **关注点：敏感性、思考广度/完整度、思考深度**